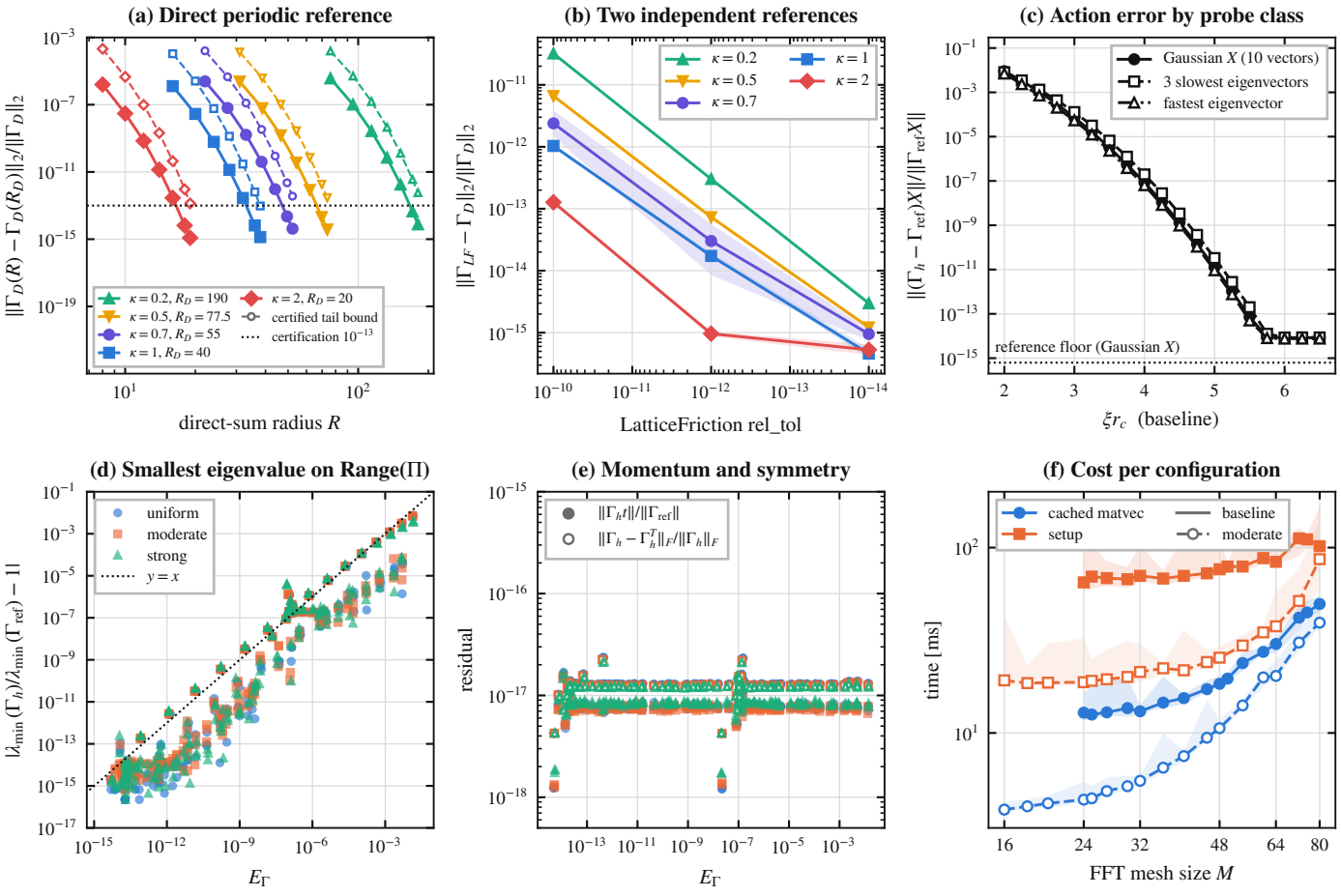

(a) Direct periodic reference
$\|\Gamma_D(R) - \Gamma_D(R_D)\|_2 / \|\Gamma_D\|_2$
direct-sum radius R
$\kappa = 0.2$, $R_D = 190$
$\kappa = 0.5$, $R_D = 77.5$
$\kappa = 0.7$, $R_D = 55$
$\kappa = 1$, $R_D = 40$
$\kappa = 2$, $R_D = 20$
certified tail bound
certification 10^{-13}
(b) Two independent references
$\|\Gamma_{LF} - \Gamma_D\|_2 / \|\Gamma_D\|_2$
LatticeFriction rel_tol
$\kappa = 0.2$
$\kappa = 0.5$
$\kappa = 0.7$
$\kappa = 1$
$\kappa = 2$
(c) Action error by probe class
$\|(\Gamma_h - \Gamma_{\mathrm{ref}})X\| / \|\Gamma_{\mathrm{ref}}X\|$
ξr_c (baseline)
Gaussian X (10 vectors)
3 slowest eigenvectors
fastest eigenvector
reference floor (Gaussian X)
(d) Smallest eigenvalue on Range(Π)
$|\lambda_{\min}(\Gamma_h)/\lambda_{\min}(\Gamma_{\mathrm{ref}}) - 1|$
E_Γ
uniform
moderate
strong
$y = x$
(e) Momentum and symmetry
residual
E_Γ
$\|\Gamma_h t\| / \|\Gamma_{\mathrm{ref}}\|$
$\|\Gamma_h - \Gamma_h^T\|_F / \|\Gamma_h\|_F$
(f) Cost per configuration
time [ms]
FFT mesh size M
cached matvec
setup
baseline
moderate